Topic 13

Sunday, October 4, 2026 4:19 PM

17 MAC_Rand...

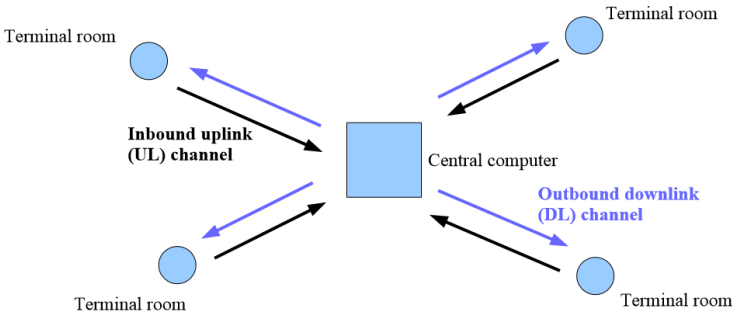
Chapter 6
Medium Access Control Protocols and Local Area Networks



Random Access

ALOHA

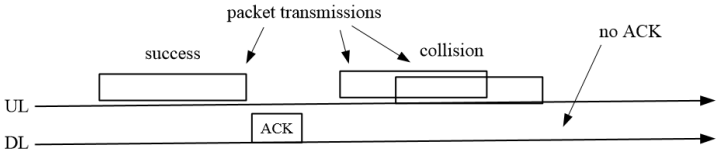
- Wireless link to provide data transfer between main campus and remote campuses at the University of Hawaii in the 1970s.
- Central computer transmits on the outbound downlink (DL) channel. Remote terminal rooms transmit on shared inbound uplink (UL) channel.



How to do medium access control?

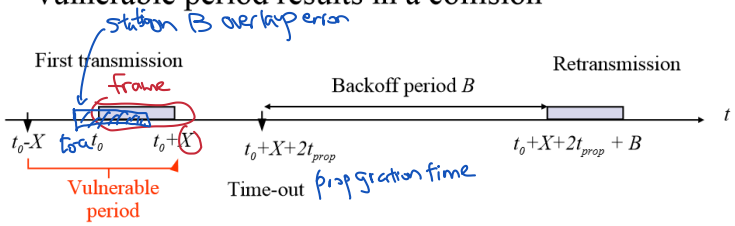
ALOHA Protocol

- A station transmits whenever it has a packet to send!
- Assume that if more than one packet overlaps in time, they interfere with each other and all are lost. This is called a collision.
- If an ACK is not received within a timeout period, then the transmitting station waits for a random backoff time (to avoid repeated collision), then retransmits the packet.



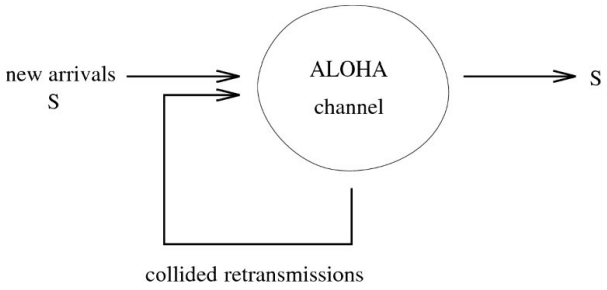
ALOHA Protocol Backoff

- Assume all packets are the same length.
- Each packet has a "vulnerable period" that is twice the transmission time of the packet.
- Any other transmission starting during the vulnerable period results in a collision

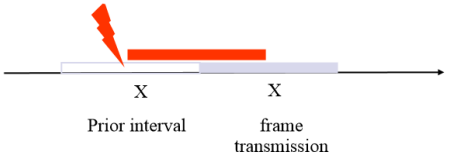


ALOHA Throughput Model

- Definitions and assumptions
  - *X* packet transmission time (assume the same for all transmissions)
  - *S*: ALOHA channel throughput (average # successful frame transmissions per *X* seconds)
  - *G*: Total channel load (average # transmission attempts per X sec.)
  - $P_{success}$: probability a packet transmission is successful

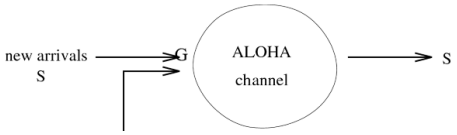


ALOHA Model



- Any transmission that begins during a packet's vulnerable period leads to collision, i.e., success if no transmission occurs during a packet's 2X second vulnerable period
- In this case, there is 1 successful packet transmission in a 2X period, i.e., 0.5 packet transmissions in each X period

- Assume a large number of stations such that the aggregate arrival process at the channel is a Poisson process, with rate G=λX.

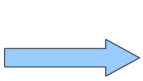




collided retransmissions

$$Pr\{n \text{ arrivals in interval of length } t\} = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

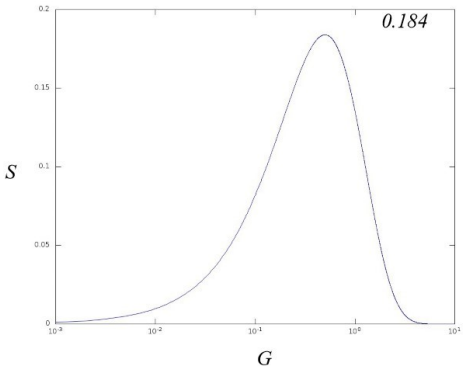
i.e., throughput $S = 0.5 \times Pr\{1 \text{ arrival in 2X interval}\} = 0.5 \times 2G\, e^{-2G}$.



$$S = G\, e^{-2G}$$

ALOHA Througput

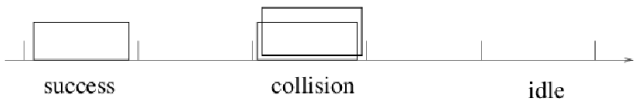
$$S = G\, e^{-2G}$$



- Collisions are means for coordinating access
- Max throughput is $\rho_{max} = 1/2e$ *(18.4%)*
- Bimodal behavior:
  Small G, S≈G
  Large G, S↓0
- Collisions can snowball and drop throughput to zero
- G may increase with number of backlogged stations.
- Holds for large numbers of stations whose individual load is low!

Slotted ALOHA

- Time is slotted in X second time slots
- Stations synchronized to slot times
- Stations transmit packets in first slot after packet arrival
- Backoff intervals in multiples of slot times



Only packets that arrive during prior X seconds collide, i.e., the vulnerable period has been cut in half.

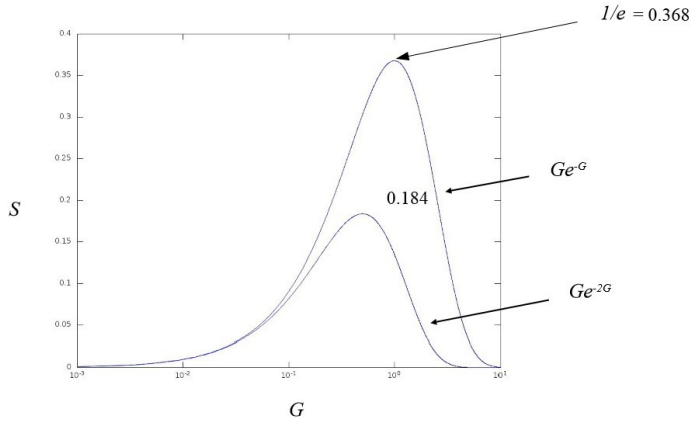
Slotted ALOHA Throughput

- Assume that transmissions on the channel occur according to a Poisson process with arrival rate G=λX.
- The probability of a successful transmission during a time slot is the probability that exactly one arrival has occurred in the previous time slot.

$$Pr\{n \text{ arrivals in interval of length } t\} = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

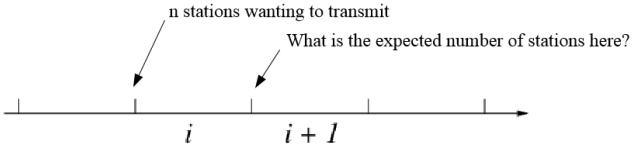
$$S = G e^{-G}$$

Slotted ALOHA Throughput



S-ALOHA Stability

- We will model the system from one time slot to another. Consider the actions of a slotted ALOHA network in going from slot i to slot $i + 1$.
- Assume that there is a large station backlog. Let n = the station backlog at start of slot i.
- Assume that each backlogged station transmits with probability q in every time slot and that new backlogged stations are being generated by a Poisson process at a rate of λ per slot.



The mean arrival rate to the channel is

$$G(n) = n q + \lambda$$

It can be shown that if q is small and n is large then the prob of k arrivals in a slot is given by the Poisson distribution,

$$Pr\{k \text{ arrivals}\} = \frac{G(n)^k e^{-G(n)}}{n!}$$

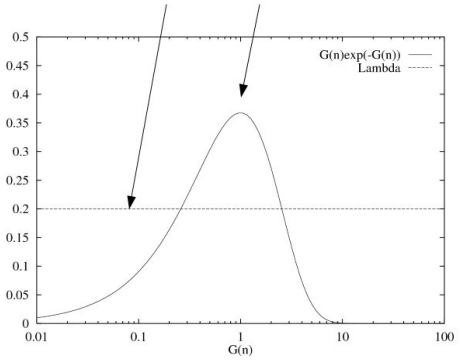
The probability of a successful transmission in a slot is therefore

$$Pr\{\text{success}\} = G(n) e^{-G(n)}$$

We define the drift $D(n)$ to be the expected change in backlog

We define the drift $D(n)$ to be the expected change in backlog from slot i to slot $i+1$, i.e., $D(n)$ = expected number of new station arrivals - expected number of successful transmissions.

$$D(n)=\lambda-G(n)e^{-G(n)}$$



ALOHA Stability

When $n\uparrow, G(n)\uparrow, D(n)\uparrow, n\uparrow, \ldots$

Infinite population ALOHA networks with fixed mean station backoffs are asymptotically unstable!

Objective:

When $n\uparrow, D(n)\downarrow$

Stabilizing ALOHA

- In a slotted ALOHA system we had that the total load impinging on the channel is given by

$$G(n)=nq+\lambda$$

- The problem is that if the number of stations is large, then n may grow to be large. If q is constant then the channel load ($G(n)$) may eventually be very high and the channel throughput, i.e.,

$$G(n)e^{-G(n)}$$

will be very low. How do we solve this problem?

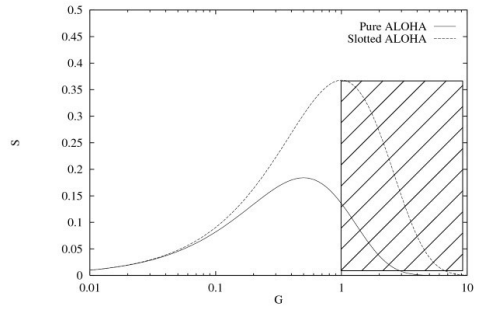
The solution is to make q a function of n, i.e.,

$$q=q(n)$$

In this case q should be made to *decrease* as n increases so that the throughput does not drop. Note that

$$G(n)e^{-G(n)}$$

has a maximum at $G(n) = 1$. Therefore when there is a large number of backlogged stations, q should be adjusted so that $G(n) = 1$. i.e., since $G(n) = q\,n + \lambda$, q must decrease as n increases so that the product of the two is constant.



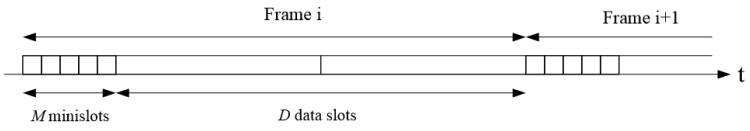
To do this exactly however, the stations must know the exact value of n, which is difficult.

Some designs use estimates of n, based on channel feedback. Others use "implicit" schemes, where q is decreased based on the number of collisions that a packet has suffered.

Dynamic TDMA with ALOHA Reservations

- This addresses the case where traffic is bursty but when a station becomes busy, it requires multiple TDMA data slots.
- These kinds of protocols are common in cable modems used by modern coxial cable based ISPs (such as Rogers, Cogeco, etc.) and in cellular air interface protocols (e.g., LTE)

Reservation ALOHA

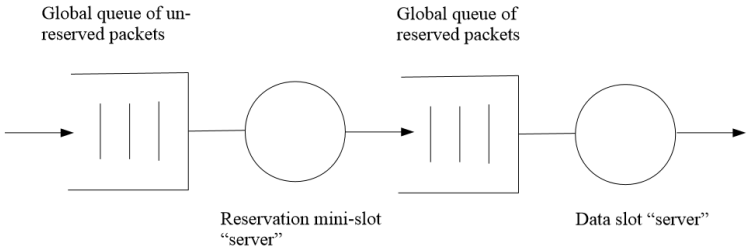


- Reservation protocol allows a large number of stations with infrequent traffic to reserve slots to transmit their frames in future cycles
- Each cycle has mini-slots allocated for making reservations
- Stations use slotted ALOHA during mini-slots to request use of data slots.
- M mini-slots of length $T_{\text{„}}$

- D data slots of length T_d

How can we compute the efficiency of R-ALOHA?

Reservation ALOHA Efficiency



System capacity is determined by the minimum of the two server capacities.

R-ALOHA Efficiency

- Throughput in mean data slots per frame is min(mean number of successful reservations/frame, number of data slots per frame).

$$\text{Max mean \# of reservations per frame } = \frac{M}{e}$$

$$\text{Max mean \# of occupied data slots per frame } = min(\frac{M}{e}, D).$$

$$\text{Efficiency} = \frac{T_d \, min(\frac{M}{e}, D)}{D T_d + M T_m}$$

PRMA

- PRMA (packet reservation multiple access) is an example of R-ALOHA but for packet voice transmission.
- Exploits voice *talkspurt detection*